Chapter 14: Optimal Policy and Stabilization bias in the New Keynesian Model

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1 Overview

We now consider optimal policy in the New Keynesian model, both under commitment and under discretion. In this chapter we focus on optimal policy while abstracting from the zero lower bound on interest rates (ZLB); in the next chapter we account for it. We start by focusing on the nature of optimal policy in the absence of shocks. An interesting finding for the optimal policy under commitment is that the government does not trade off inflation and output, as was the case for the standard Phillips curve, even if the form of the Phillips curve may suggest that such trade-offs are there. Instead, in the absence of shocks, the optimal long run inflation rate is perfect price stability, a result first shown in Woodford (2003). We then show that the inflation bias emerges, yet again, as a key dynamic inconsistency problem. As in the case of the New Classical Phillips curve, if the government takes expectations as given, it will be tempted to create too much inflation in equilibrium. We then move on to analyzing the reaction of shocks both under commitment and discretion, uncovering a different dynamic inconsistency problem in the case of cost-push shocks. This problem was first shown by Clarida, Galí and Gertler (1999), who term it the stabilization bias. The next chapter then shows a different dynamic inconsistency problem that arises if the shocks are large enough that the ZLB becomes a constraint: the deflation bias, first shown by Eggertsson (2006).

2 The Case for Price Stability in the New Keynesian Model

In the last chapter I showed that you can summarize the New Keynesian model with an objective function:

$$E_0 \sum_{t=0}^{\infty} \beta^t L_t \tag{1}$$

where

$$L_t = \frac{1}{2}\pi_s^2 + \frac{1}{2}\lambda_x (x_s - x^*)^2 \tag{2}$$

and the IS and AS curves are, respectively:

$$x_t = E_t x_{t+1} - \sigma_c(\hat{\imath}_t - E_t \pi_{t+1} - \hat{r}_t^e) \tag{3}$$

$$\pi_t = \kappa x_t + \beta E_t \pi_{t+1} + \omega \hat{\mu}_t \tag{4}$$

where x_t is the deviation of output from the efficient output level. We ignore the ZLB for now, so that, as in our earlier analysis, we can omit the IS curve when characterize the optimal plan, as the IS curve simply backs out the interest rate the central bank needs to target to implement the optimal paths for x_t and π_t.

The optimal plan under commitment can be found using the method of Lagrange. The Lagrangian for this problem is given by:

$$L_0 = E_0 \sum_{s=t}^{\infty} \beta^{s-t} \{\frac{1}{2}\pi_s^2 + \frac{1}{2}\lambda_x (x_s - x^*)^2 + \phi_s \{\pi_s - \kappa x_s - \beta \pi_{s+1} - \omega \hat{\mu}_t\}\} \tag{5}$$

The first-order conditions for $t = 0$ are:

$$\frac{\partial L_0}{\partial \pi_0} = \pi_0 + \phi_0 = 0 \tag{6}$$

$$\frac{\partial L_0}{\partial x_0} = \lambda (x_0 - x^*) - \kappa \phi_0 = 0 \tag{7}$$

and for $t > 0$:

$$\frac{\partial L_0}{\partial \pi_t} = \pi_t + \phi_t - \phi_{t-1} = 0 \tag{8}$$

$$\frac{\partial L_0}{\partial x_t} = \lambda (x_t - x^*) - \kappa \phi_t = 0 \tag{9}$$

An equilibrium under the optimal commitment at date $t \geq 0$ is thus a collection of stochastic processes for $\{\pi_t, x_t, \phi_t\}$ that solves equations (4), (6), (7), (8), (9) for an exogenous process $\{\mu_t\}$. This gives us the next proposition, which establishes that under the optimal policy the central bank will not trade off inflation against the output gap.

Proposition 1. *In the long run in the New Keynesian model, optimal policy under commitment does not trade off inflation and the output gap so that* $\pi_t \to 0$.

To prove this proposition combine the first-order conditions with the AS equation to obtain:

$$\beta E_t \phi_{t+1} - (1 + \beta + \kappa^2/\lambda)\phi_t + \phi_{t-1} = \kappa x^* + \omega \hat{\mu}_t \tag{10}$$

and using the method of Blanchard and Kahn outlined in Chapter 3, we see that that this difference equation has to have one eigenvalue outside of the unit circle and the other inside of it (see Example 1 in Chapter 3 that goes through the calculations step by step). The eigenvalues solve the polynomial $\beta\gamma^2 - (1 + \beta + \kappa^2/\lambda)\gamma + 1$ where $0 < \gamma_1 < 1 < \gamma_2$.

Focusing on the deterministic component (i.e. setting the shock $\hat{\mu}_t = 0$), we have a unique bounded solution given by:

$$\phi_t = -\lambda/\kappa x^*(1 - \gamma_1^{t+1})$$

$$\pi_t = (1 - \gamma_1)\frac{\lambda}{\kappa} x^* \gamma_1^t$$

The time path for inflation is shown in figure 1, taken from Woodford (2003). The solution is denoted "zero-optimal." The optimal solution under commitment includes initial inflation which then converges to 0 as time goes by. Of particular interest for now are the contrasts to the optimal policy for the standard New Keynesian model. In that model, the government exploited the long-run trade-off between inflation and output. Interestingly, the government chooses not to do so in this case, even if there is such a trade-off in steady state. To see this consider the steady state (i.e. constant solution in absence of shocks) of equation (4), and denote it by π^{ss}, x^{ss}. The trade-off is then given by

$$x^{ss} = \pi^{ss}\frac{1 - \beta}{\kappa} \tag{11}$$

and hence the output gap is increasing in inflation. Notwithstanding this, and perhaps somewhat surprisingly, the proposition above reveals that, in the long run, the optimal policy mandates full price stability.

That inflation converges to zero can be understood by integrating the AS equation (4) forward to yield:

$$\pi_0 = \kappa E_0 \sum_{t=0}^{\infty} \beta^t x_t \tag{12}$$

Let us now rewrite the loss function (2) as:

$$L_t = \frac{1}{2}\pi_t^2 - \lambda_x x^* x_t + \frac{1}{2}\lambda_x x_t^2 \tag{13}$$

We can now use (12) to rewrite the loss function once more as:

$$E_0 \sum_{t=0}^{\infty} \beta^t L_t = -\kappa\lambda_x x^* \pi_0 + E_0 \sum_{t=0}^{\infty} \beta^t [\frac{1}{2}\pi_t^2 + \frac{1}{2}\lambda_x x_t^2] \tag{14}$$

Recall that the optimal commitment is made at time 0. Observe that according to this expression, only inflation in period 0 interacts with x^*. Committing to inflation in any later period has no beneficial effect in terms of welfare. To obtain some intuition, think about the effect of increasing π_t by 1 percent. This gain increases the output gap by κ^{-1} percent in period t. Note, however that π_t will similarly shift the output gap x_{t-1} at time $t-1$. In particular it will reduce the output gap by β percent times κ^{-1}, as the expectation of inflation appears in the Phillips curve in the previous period. From the standpoint of the discounted loss criterion, the costs are weighted more heavily than the benefit by a factor of β^{-1}. Hence they exactly offset each other, and accordingly full price stability is optimal.

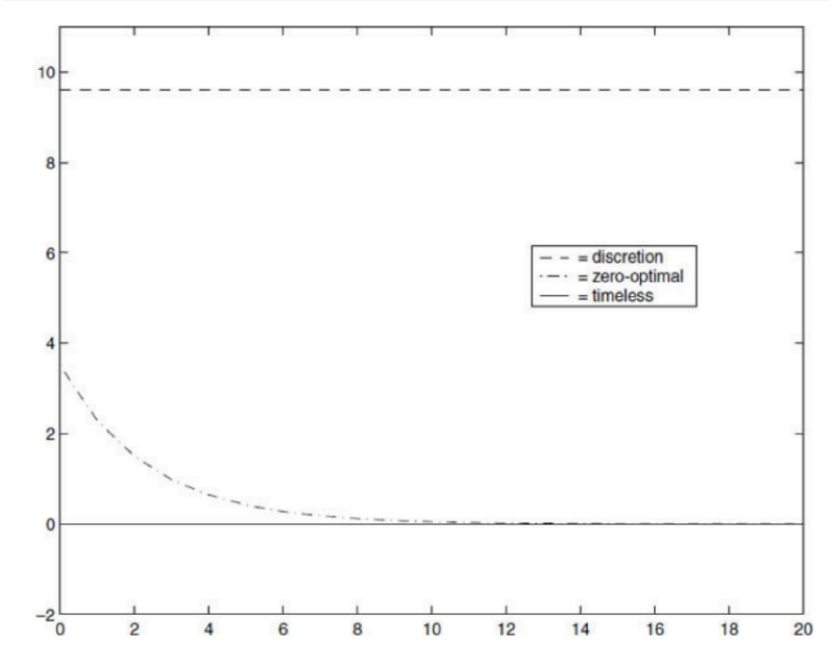


Figure 1: Optimal Policy in the NK Model

3 The Inflation Bias

Now we consider optimal policy under discretion. The definition of discretion is that the government is unable to commit to future policy, hence it must take expectations of the private sector as given. These assumptions, meanwhile, are formed under rational expectations. One way of stating this is that the decision of the households and the government can only depend on the physical state variables of the model, which are $\hat{r}_t^e$ and $\hat{\mu}_t$. Hence we have $E_t\pi_{t+1} = \bar{\pi}(\hat{r}_t^e, \hat{\mu}_t)$, $E_t x_{t+1} = \bar{x}(\hat{r}_t^e, \hat{\mu}_t)$, i.e. the expectations are a function of the exogenous shocks but independent of the policy choice at time t. The problem is completely static, so we can consider the following Lagrangian:

$$L_t = \frac{1}{2}\pi_t^2 + \frac{1}{2}\lambda_x(x_t - x^*)^2 + \phi_t(\pi_t - \kappa x_t - \beta\bar{\pi} - \omega\hat{\mu}_t) \tag{15}$$

$$\frac{\partial L_t}{\partial \pi_t} = \pi_t + \phi_t \tag{16}$$

$$\frac{\partial L_t}{\partial x_t} = \lambda_x(x_t - x^*) - \phi_t\kappa = 0 \tag{17}$$

Combine the two first-order conditions to yield:

$$\pi_t + \frac{\lambda_x}{\kappa}(x_t - x^*) = 0$$

This equation can now be combined with the AS equation, which is then forwarded, to yield:

$$\pi_t = \frac{\kappa\lambda_x}{(1-\beta)\kappa + \kappa^2}x^* + \omega\sum_{s=0}^{\infty}(\beta\frac{\lambda_x}{\lambda_x + \kappa^2})^t E_t\hat{\mu}_{t+s}$$

Considering the solution in the absence of shocks (i.e. steady state) yields the next proposition:

Proposition 2. ***The Inflation bias.*** *Under optimal policy under discretion in the New Keynesian model, steady state inflation is positive and given by* $\pi^{ss} = \frac{\kappa\lambda_x}{(1-\beta)\kappa+\kappa^2}x^*$.

Thus, just as in the case of the New Classical Phillips curve, a discretionary government will produce inefficiently high inflation. The discretionary steady state inflation is shown in figure 1, from a numerical example in Woodford (2003).

4 The Timeless Perspective

Imagine you have a job at a central bank which involves presenting the president of your bank with a plan for optimal policy every four periods. Suppose you believe that the New Keyenesian model describes the truth. What should your plan look like?

One strategy would be to follow the discretionary model. This strategy would have the advantage that your advice would be consistent. Consult figure 1 and tell you president to aim for about 10 percent inflation.

But this is hardly good advice, as the zero-optimal plan is clearly much preferable. But should you propose that plan to your superior instead? Looking at the time path, you would be suggesting 4 percent inflation at time 0 that then progressively goes to steady state. But what about your next planned meeting four periods later? Would you stick to the period 0 plan and recommend progressively going to zero, thus perhaps recommending 1 percent inflation in that meeting and going further down in the next? But what is so special about period 0? And how do you decide which of your meetings corresponds to period 0? Presumably other staff members had given advice before you have. Should you recompute the optimal plan at time 4, thus recommending 4 percent inflation again? Clearly that does not work either, because if people anticipate this behavior, you will soon find yourself in the purely discretionary equilibrium.

The problem described above, about how one should formulate policy advice and take account of "period 0" is a real one faced by all economists who have to use models to give advice. Perhaps somewhat surprisingly, the literature still has not come up with a satisfactory solution to it.

The most promising and prominent solution to this problem is the so-called "timeless" perspective from Woodford (2003). Roughly speaking, it boils down to imagining that the advice you give at any time should be formulated *as if* the advice were given far back in the past, so that you keep from exploiting the choice of "period zero."

5 The Stabilization Bias

Even assuming $x^* = 0$ there is still a dynamic inconsistency problem. To see this, compare the combined FOC under the optimal commitment (18) to the combined FOC under discretion (19), where we make the assumption that $x^* = 0$:

$$\pi_t + \frac{\lambda_x}{\kappa}(x_t - x_{t-1}) = 0 \tag{18}$$

$$\pi_t + \frac{\lambda_x}{\kappa} x_t = 0 \tag{19}$$

We can solve the model by combining the conditions above with the AS equation. The solution is shown in figure 2, from a numerical example in Woodford (2003). The assumption is that the cost-push shock, $\hat{\mu}_t$, is iid. The figure shows the response of output inflation and the price level to a one time cost-push shock.

Under discretion, shown by the dashed line, the policy maker trades off inflation and the output gap in period 0. In period 1, however, it fully stabilizes the output gap and the price level, as this represents the first-best outcome.

In contrast, under the optimal policy commitment, the policy maker will "smooth out" the shock, by spreading the output gap over several periods. Moreover, while there is inflation in period 0, the policy maker undershoots the inflation target in period 1 and only gradually brings it back to its steady state level.

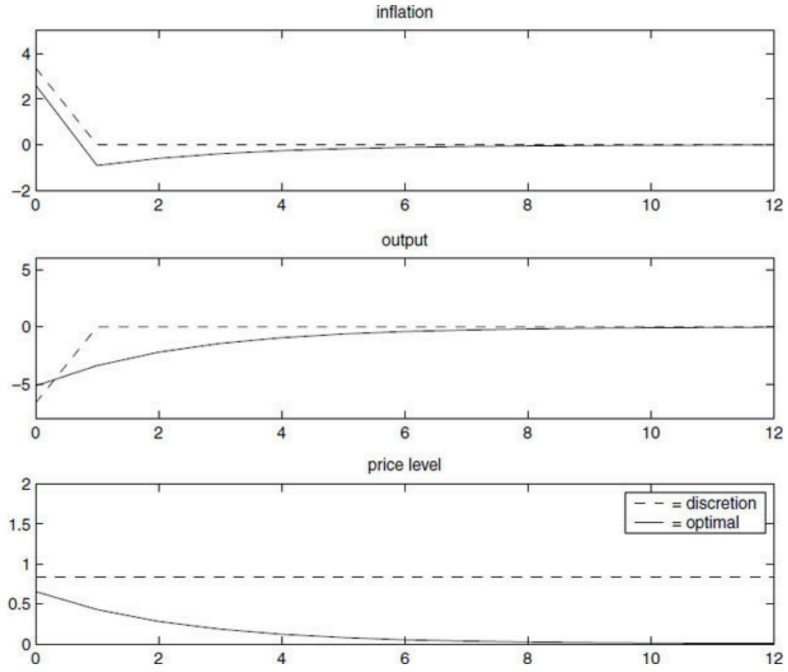


Figure 7.3 *Optimal responses to a transitory cost-push shock compared with equilibrium responses under discretionary policy.*

Figure 2: Optimal response under discretion and commitment to a cost-push shock

6 Policy Implementation

There is a rich literature on how you can implement optimal policy, for example via different policy rules.

One example is instrument rule, such as the Taylor rule.

A second is a "targeting rule," under which the policy maker commits to satisfying a certain targeting criterion such as:

$$\pi_t + \frac{\lambda_x}{\kappa}(x_t - x_{t-1}) = 0 \tag{20}$$